RS57 Yield, Efficiency & Cross-Section Comparison

Existing vs. Latest Results

October 28, 2026

Single Differential Cross-Section Formalism ($\mathrm{d}\sigma/\mathrm{d}p_T$)

LH_2 (pp) Single Differential Cross Section:

$$\frac{\mathrm{d}\sigma_{pp}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

LD_2 (pd) Single Differential Cross Section:

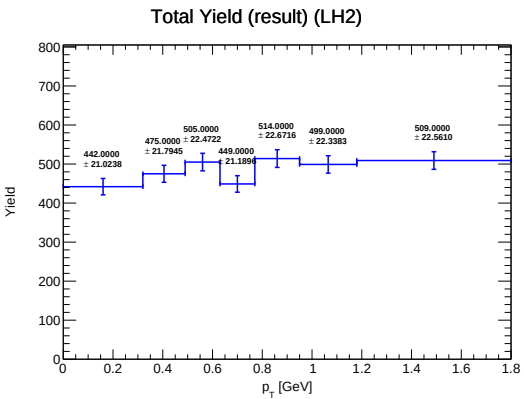
$$\begin{aligned}\frac{\mathrm{d}\sigma_{pd}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\Bigg(&\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \\ &- \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\Bigg)\end{aligned} \tag{2}$$

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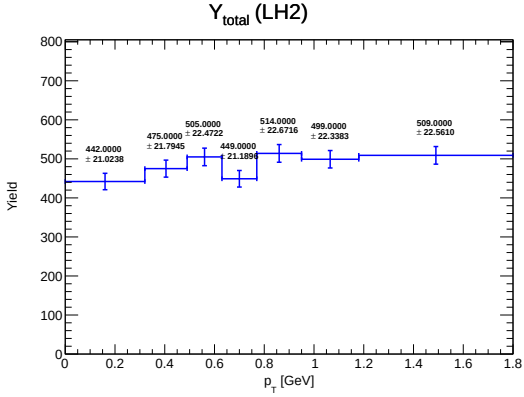
1. Yield Distributions
2. Efficiency Distributions
3. Corrected & Subtracted Yields
4. Cross Sections & Ratios
5. Systematic Uncertainties

LH_2 Target: Total Yield (Y_total_LH2.pdf)

Existing Results

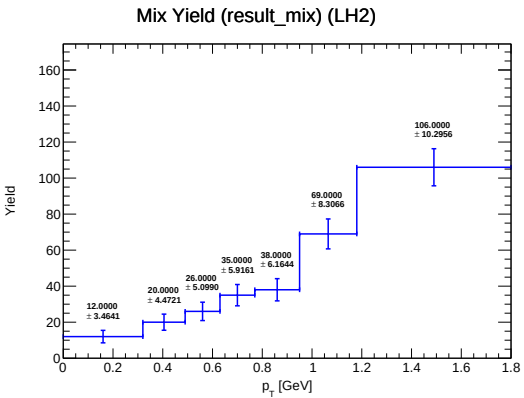


Latest Results

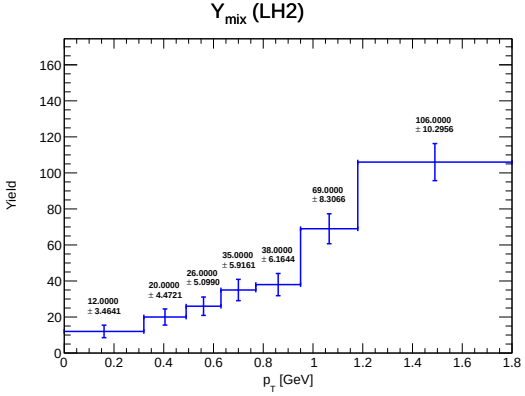


LH_2 Target: Mixed Yield (Y_mix_LH2.pdf)

Existing Results

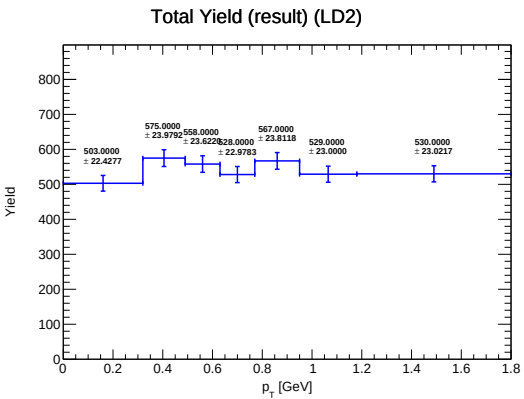


Latest Results

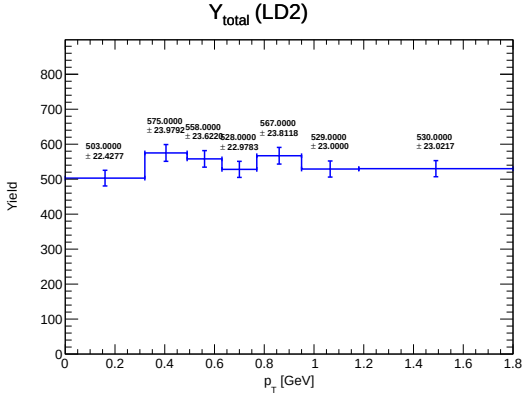


LD_2 Target: Total Yield (Y_total_LD2.pdf)

Existing Results

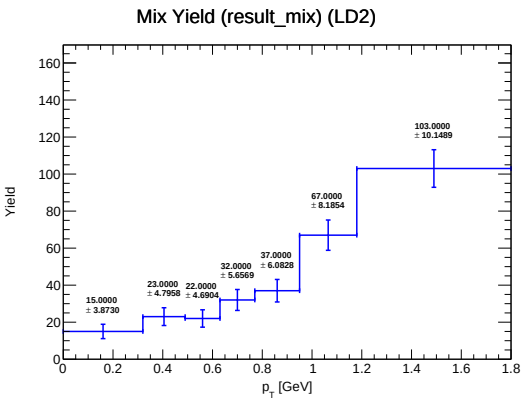


Latest Results

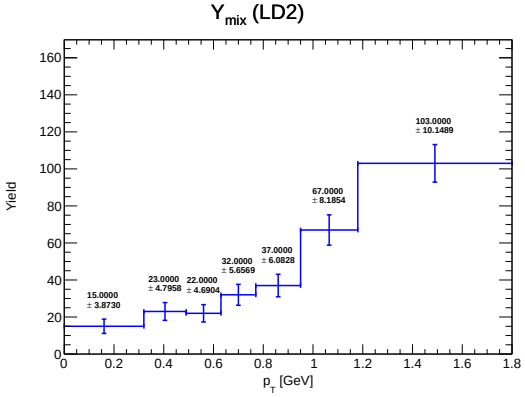


LD_2 Target: Mixed Yield (`Y_mix_LD2.pdf`)

Existing Results

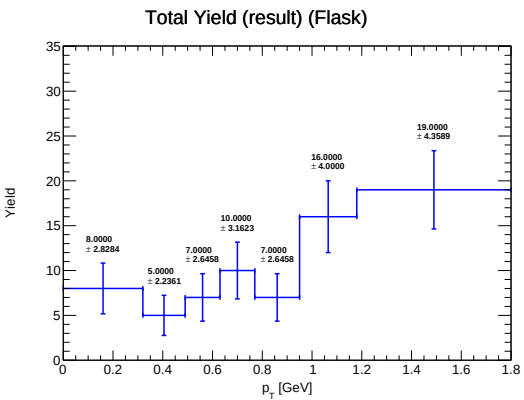


Latest Results

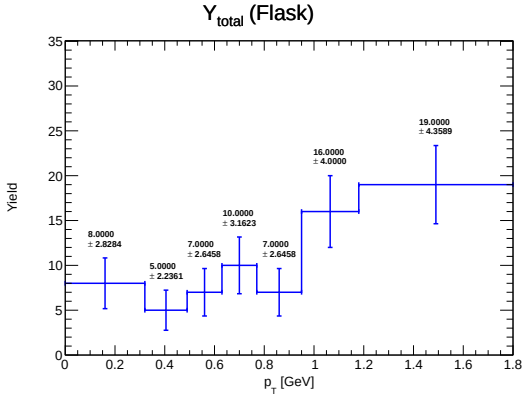


Empty Flask: Total Yield (Y_total_Flask.pdf)

Existing Results

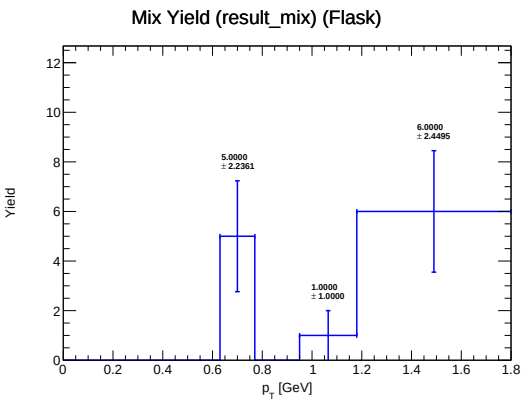


Latest Results

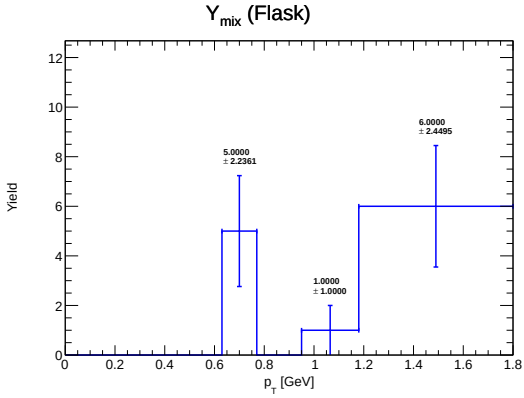


Empty Flask: Mixed Yield (Y_mix_Flask.pdf)

Existing Results

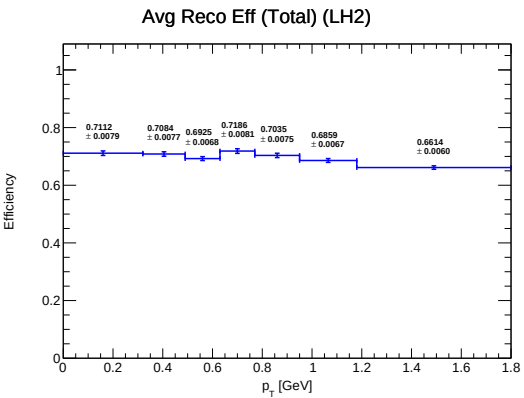


Latest Results

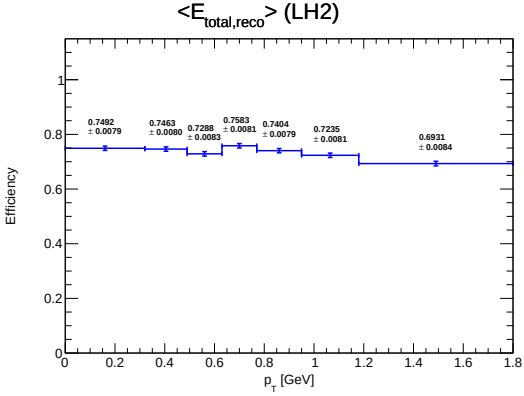


LH_2 Target: Total Reco Efficiency (E_total_reco_LH2.pdf)

Existing Results

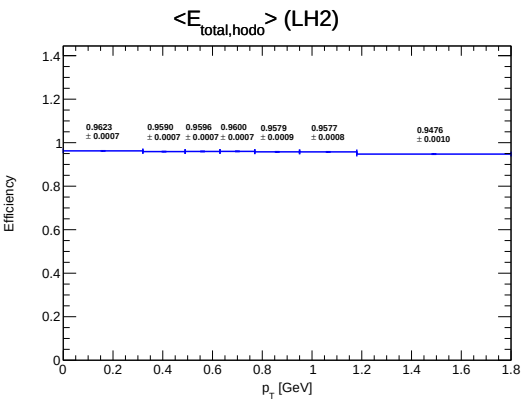


Latest Results

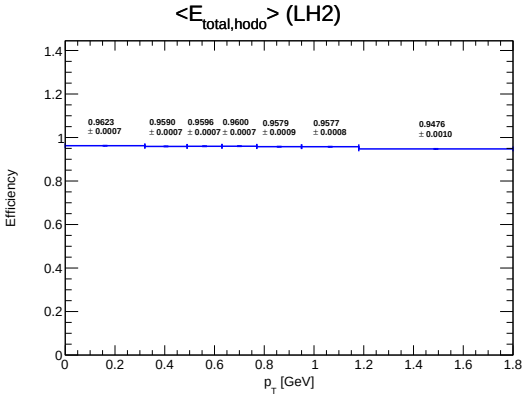


LH_2 Target: Total Hodo Efficiency (E_total_hodo_LH2.pdf)

Existing Results

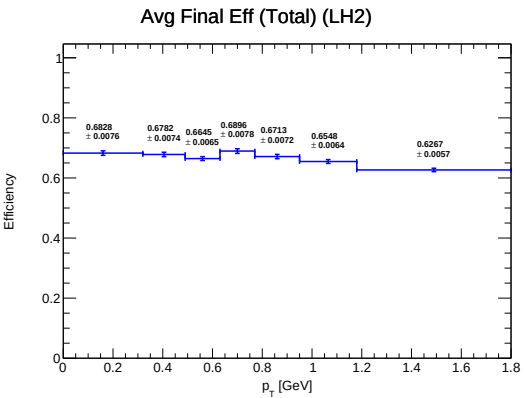


Latest Results

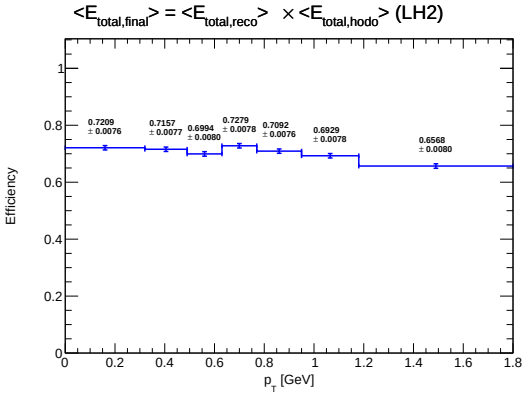


LH_2 Target: Total Final Efficiency (E_total_final_LH2.pdf)

Existing Results

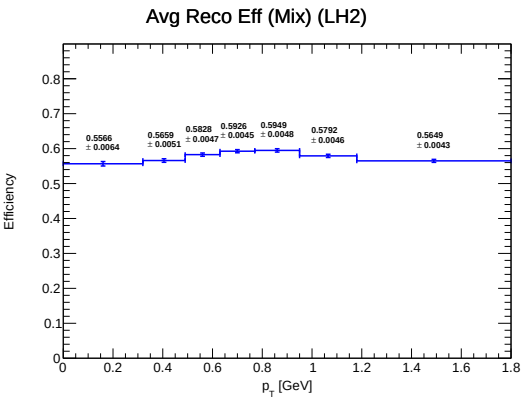


Latest Results

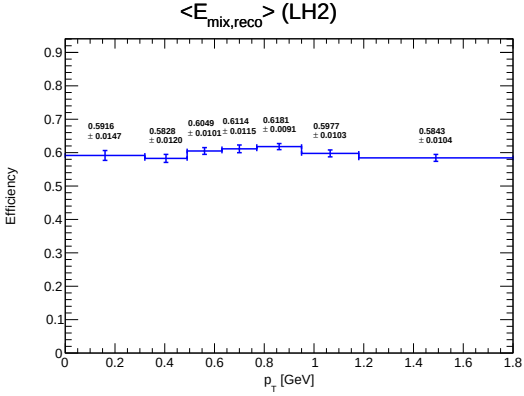


LH_2 Target: Mixed Reco Efficiency (E_mix_reco_LH2.pdf)

Existing Results

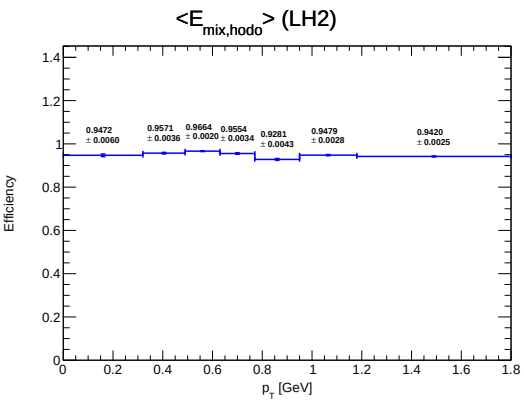


Latest Results

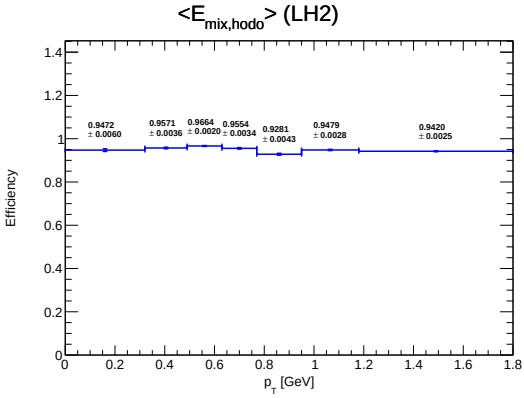


LH_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LH2.pdf)

Existing Results

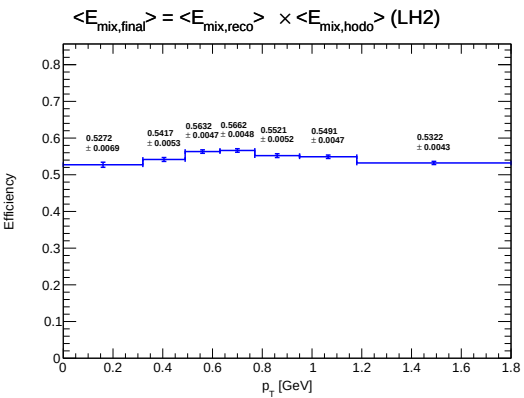


Latest Results

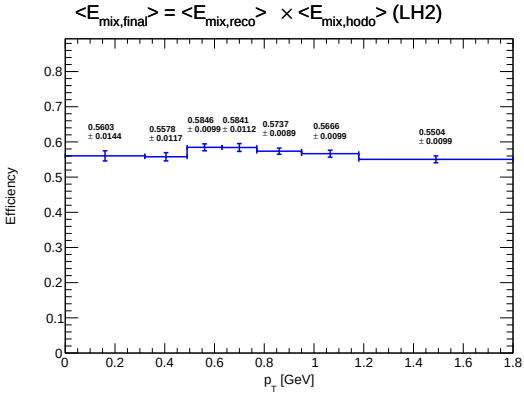


LH_2 Target: Mixed Final Efficiency (E_mix_final_LH2.pdf)

Existing Results

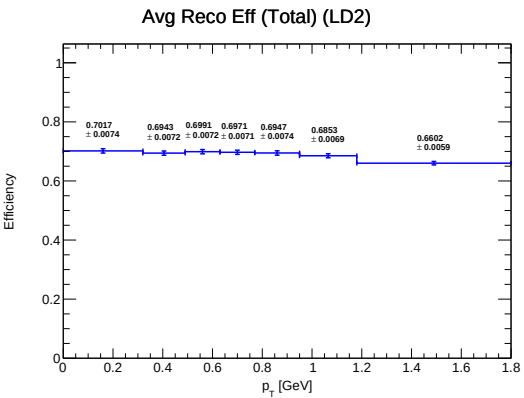


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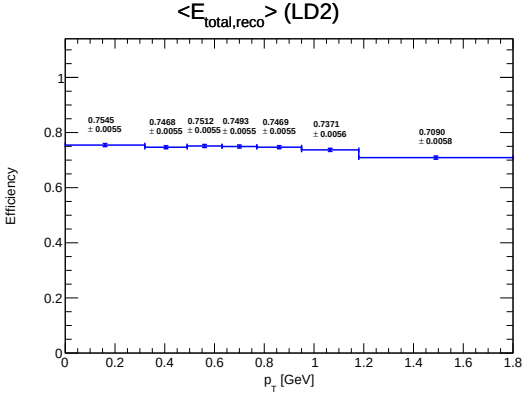


LD_2 Target: Total Reco Efficiency (E_total_reco_LD2.pdf)

Existing Results

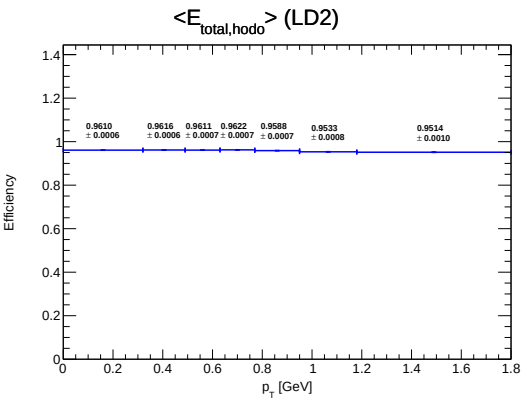


Latest Results

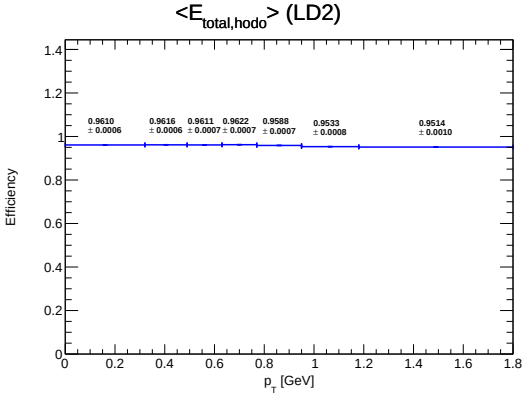


LD_2 Target: Total Hodo Efficiency (E_total_hodo_LD2.pdf)

Existing Results

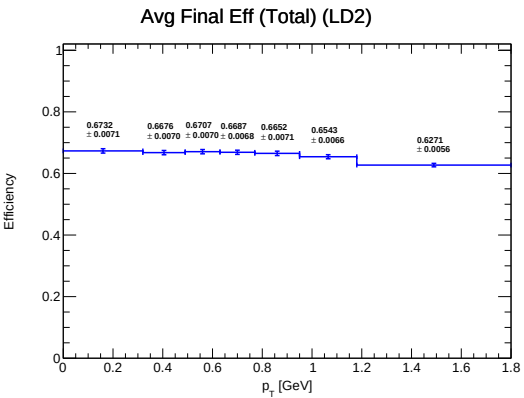


Latest Results

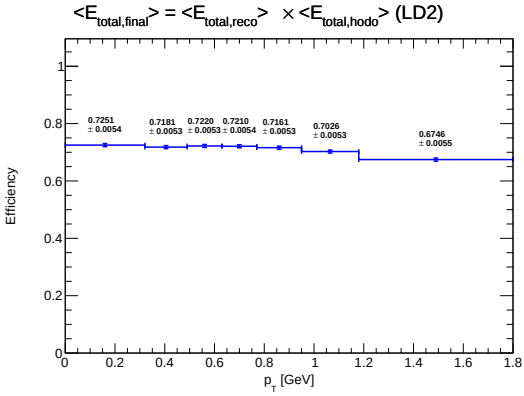


LD_2 Target: Total Final Efficiency (E_total_final_LD2.pdf)

Existing Results

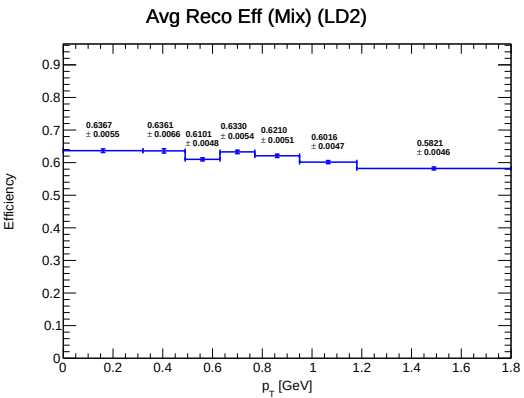


Latest Results

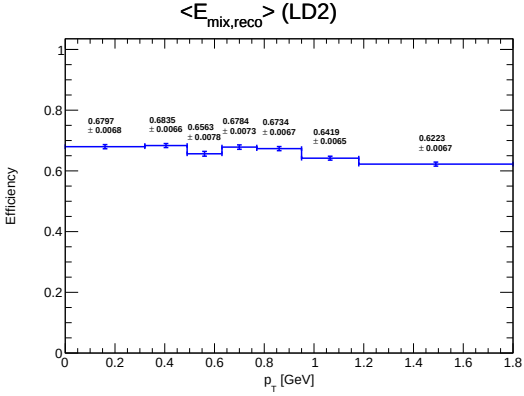


LD_2 Target: Mixed Reco Efficiency (E_mix_reco_LD2.pdf)

Existing Results

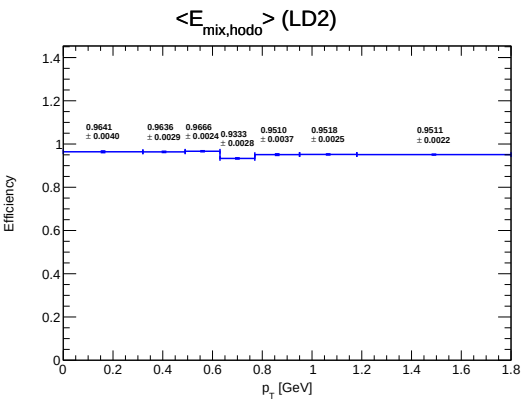


Latest Results

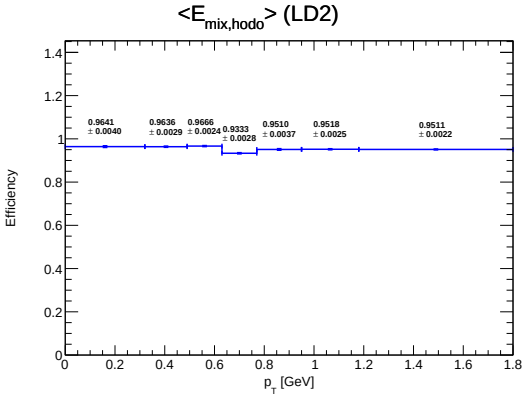


LD_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LD2.pdf)

Existing Results

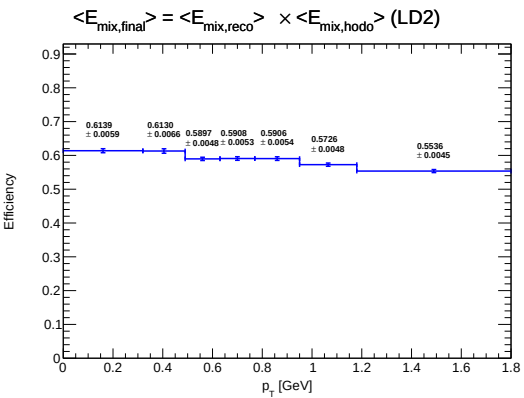


Latest Results

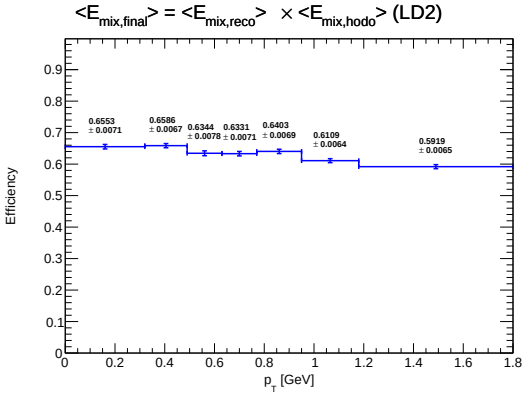


LD_2 Target: Mixed Final Efficiency (E_mix_final_LD2.pdf)

Existing Results

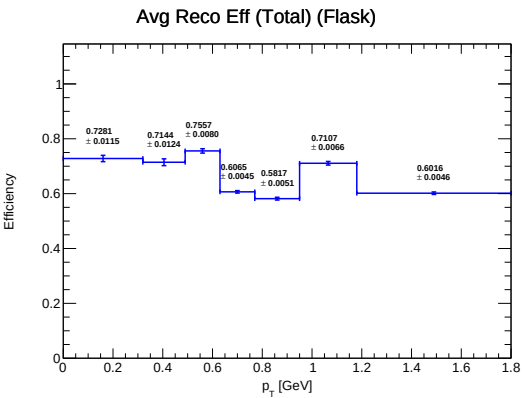


Latest Results

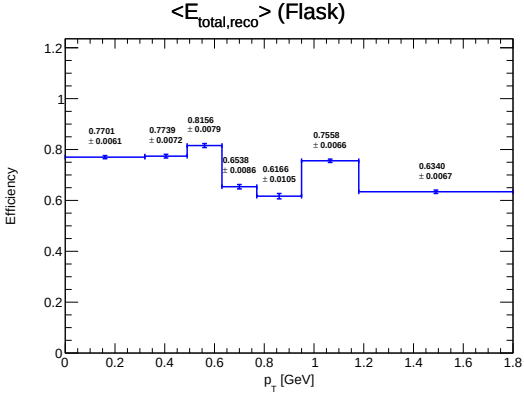


Empty Flask: Total Reco Efficiency (E_total_reco_Flask.pdf)

Existing Results

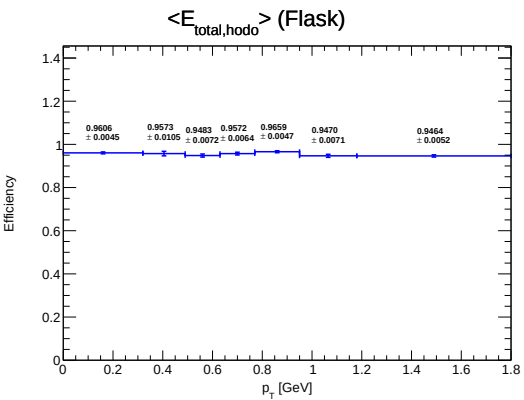


Latest Results

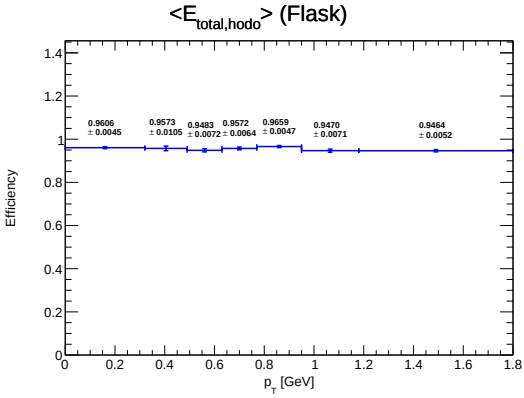


Empty Flask: Total Hodo Efficiency (E_total_hodo_Flask.pdf)

Existing Results

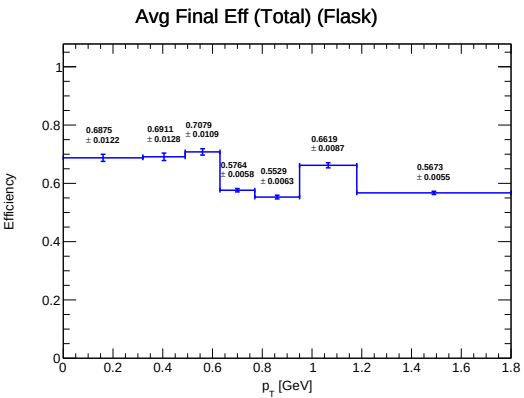


Latest Results

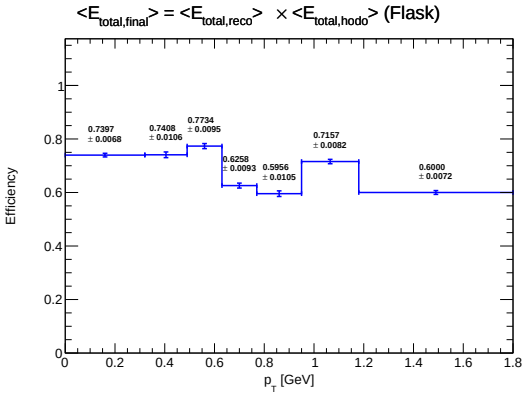


Empty Flask: Total Final Efficiency (E_total_final_Flask.pdf)

Existing Results

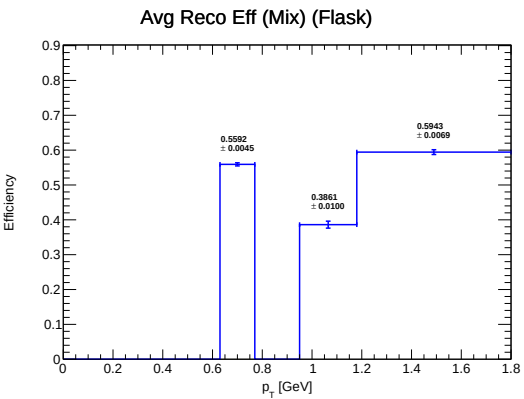


Latest Results

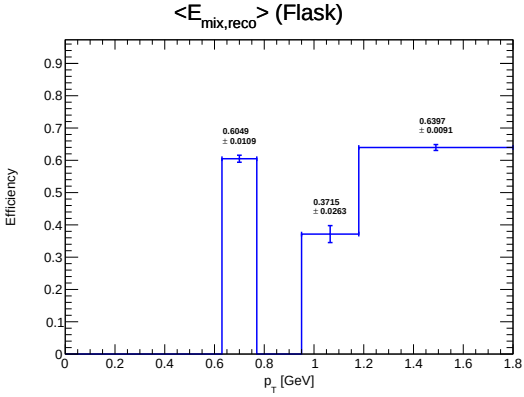


Empty Flask: Mixed Reco Efficiency (E_mix_reco_Flask.pdf)

Existing Results

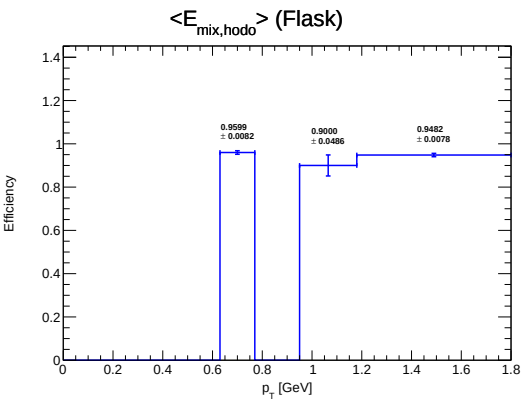


Latest Results

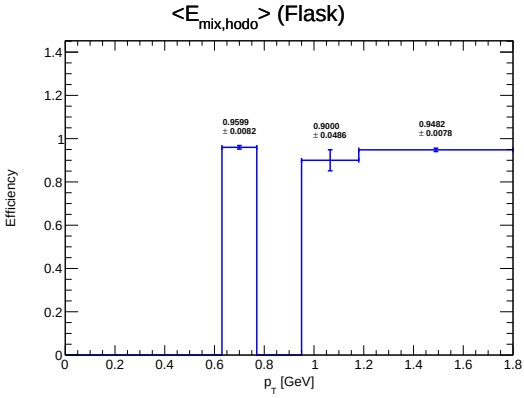


Empty Flask: Mixed Hodo Efficiency (E_mix_hodo_Flask.pdf)

Existing Results

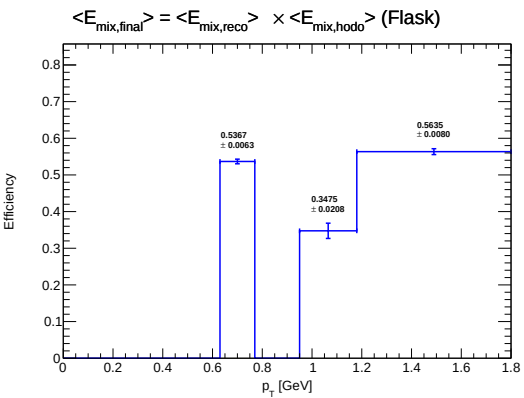


Latest Results

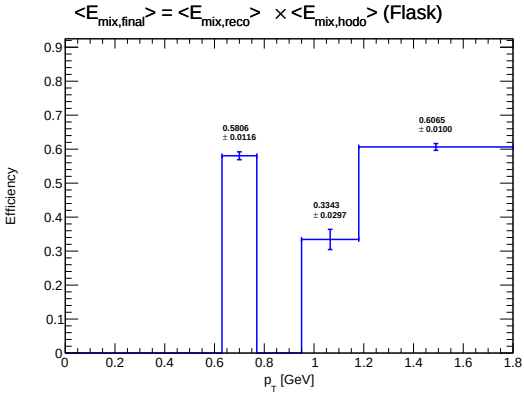


Empty Flask: Mixed Final Efficiency (E_mix_final_Flask.pdf)

Existing Results

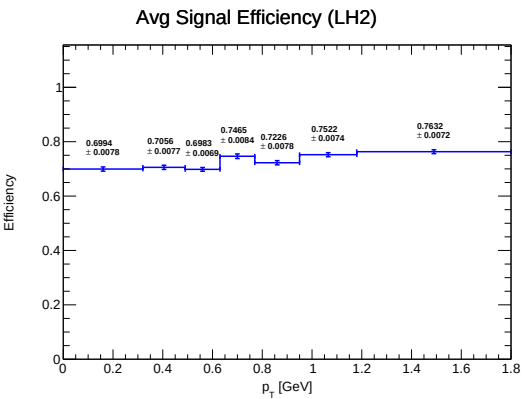


Latest Results

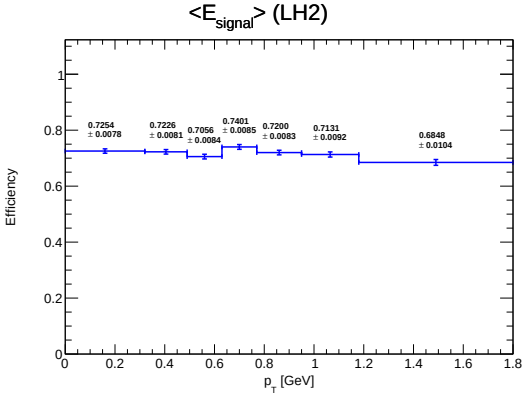


LH_2 Target: Final Signal Efficiency (`E_final_signal_LH2.pdf`)

Existing Results

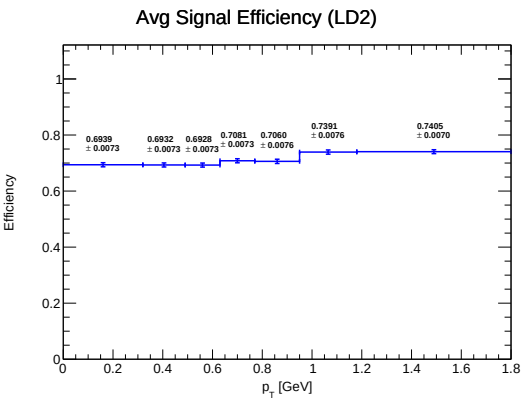


Latest Results

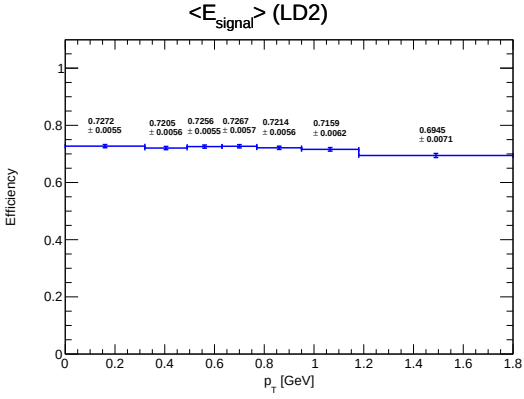


LD_2 Target: Final Signal Efficiency (`E_final_signal_LD2.pdf`)

Existing Results

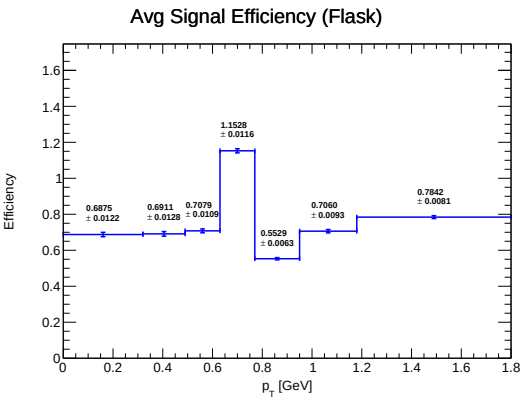


Latest Results

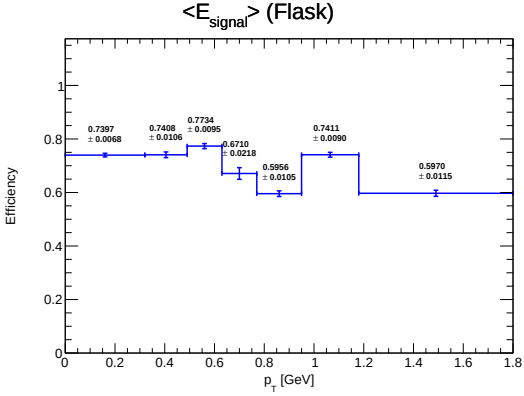


Empty Flask: Final Signal Efficiency (E_final_signal_Flask.pdf)

Existing Results

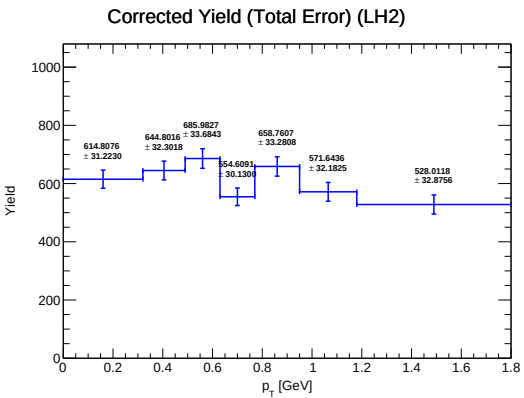


Latest Results

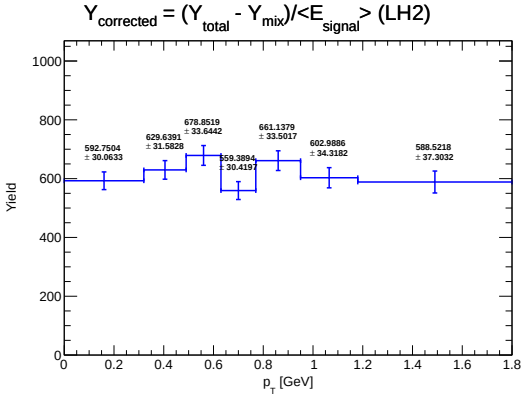


LH_2 Target: Corrected Yield (Y_corrected_LH2.pdf)

Existing Results

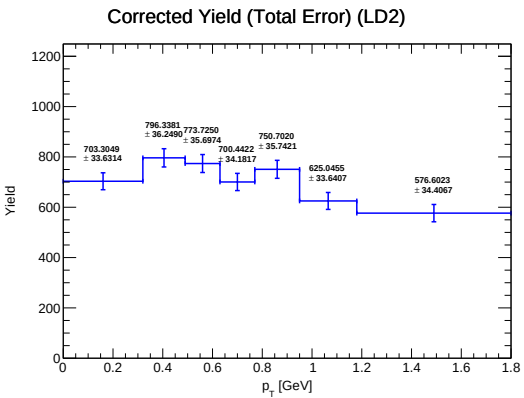


Latest Results

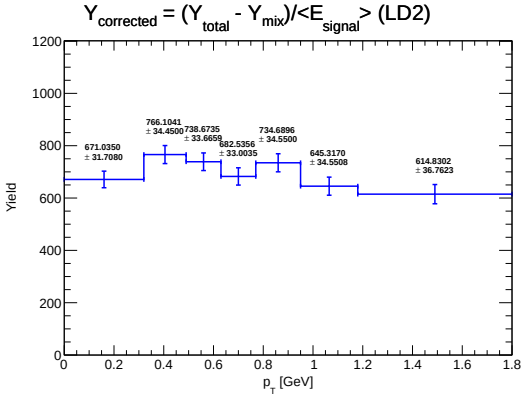


LD_2 Target: Corrected Yield (Y_corrected_LD2.pdf)

Existing Results

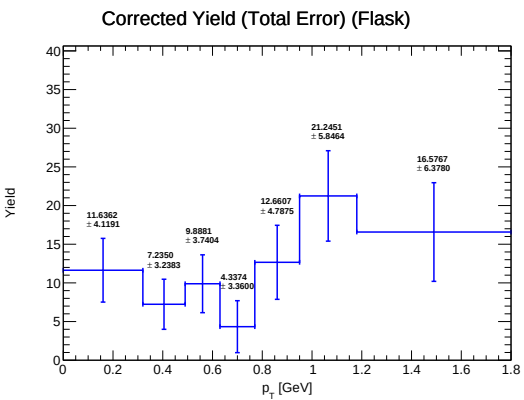


Latest Results

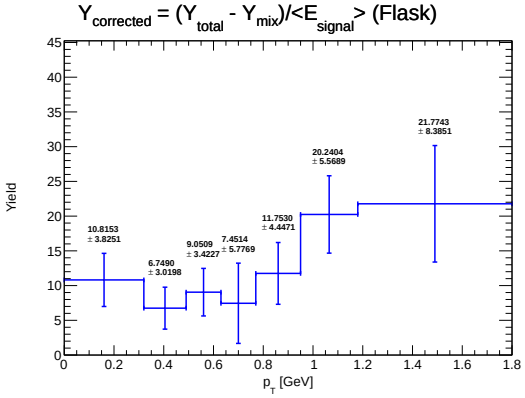


Empty Flask: Corrected Yield (Y_corrected_Flask.pdf)

Existing Results

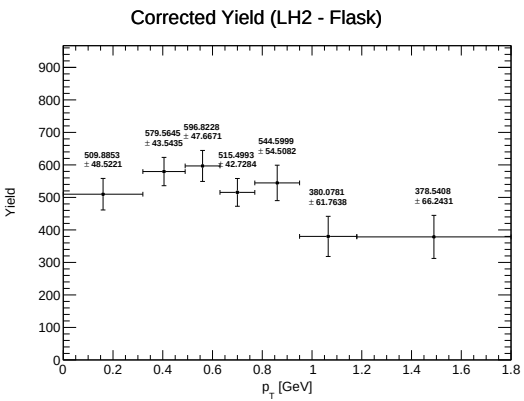


Latest Results

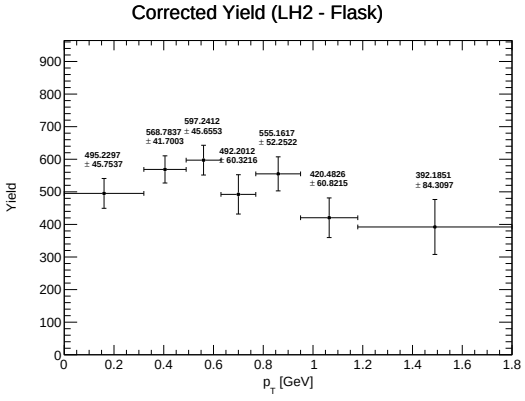


LH_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LH2.pdf)

Existing Results

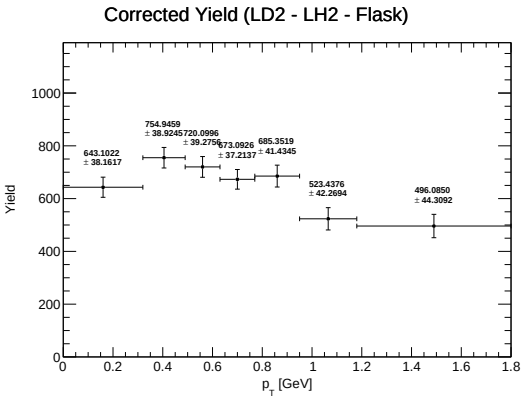


Latest Results

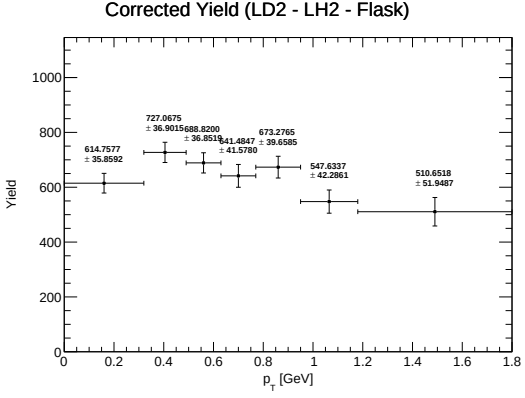


LD_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LD2.pdf)

Existing Results

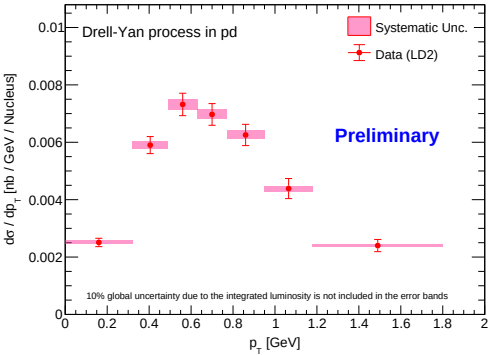


Latest Results

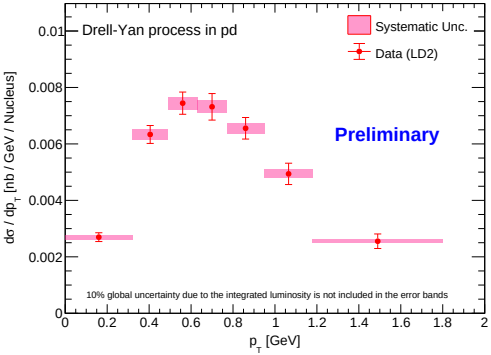


LD_2 Cross Section vs. p_T (CrossSection_LD2_geom_vs_pT.pdf)

Existing Results

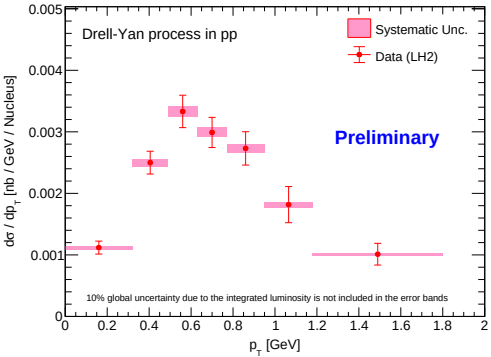


Latest Results

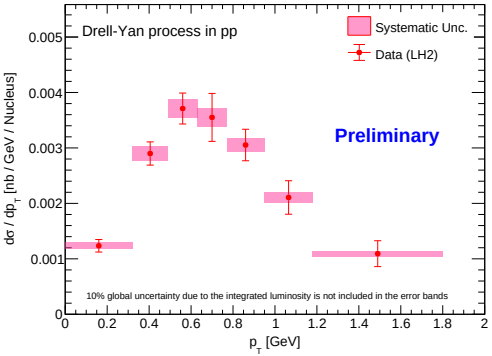


LH_2 Cross Section vs. p_T (CrossSection_LH2_geom_vs_pT.pdf)

Existing Results

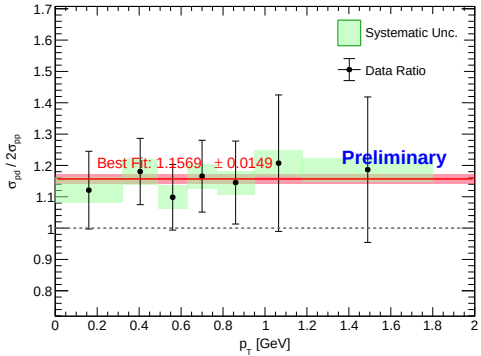


Latest Results

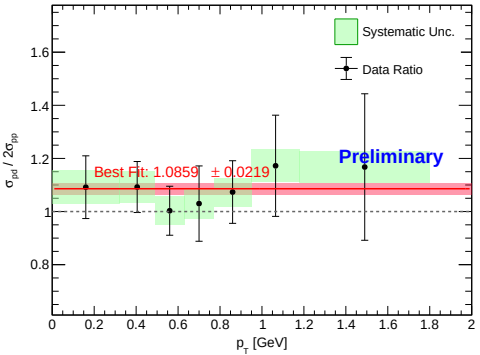


Cross Section Ratio $\sigma_{pd}/2\sigma_{pp}$ vs. p_T (CrossSection_Ratio_pd_2pp_vs_pT_geom.pdf)

Existing Results

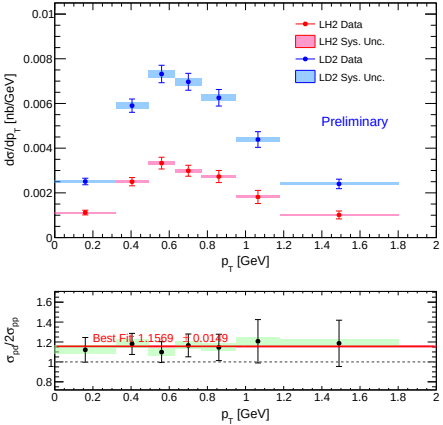


Latest Results

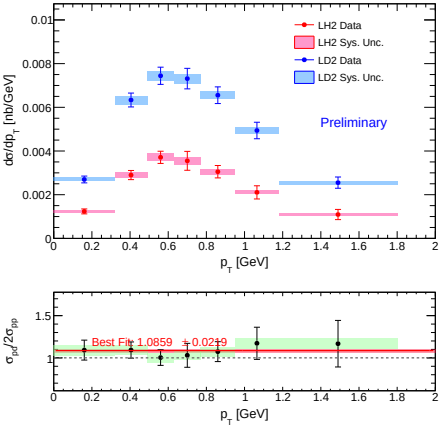


Combined Cross Section & Ratio vs. p_T (Combined_XSec_Ratio_vs_pT_geom.pdf)

Existing Results



Latest Results



LH_2 Target: Systematic Components (Table_Systematics_LH2.tex)

Latest Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.017079	0.000772	0.045198	0.001829	0.000453	0.247808	495.2297	6.4738	0.013072
[0.32, 0.49)	0.015731	0.000676	0.042960	0.002900	0.000720	0.248166	568.7837	7.0946	0.012473
[0.49, 0.63)	0.015673	0.000671	0.042812	0.003874	0.000957	0.247144	597.2412	8.1488	0.013644
[0.63, 0.77)	0.013498	0.000600	0.044437	0.003034	0.001294	0.426602	492.2012	6.8197	0.013855
[0.77, 0.95)	0.013773	0.000532	0.038626	0.003310	0.000768	0.232102	555.1617	7.8084	0.014065
[0.95, 1.18)	0.011832	0.000456	0.038521	0.004258	0.000933	0.219190	420.4826	8.1045	0.019274
[1.18, 1.80)	0.007889	0.000280	0.035430	0.003692	0.000794	0.215045	392.1851	9.7044	0.024744

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.019414	0.000441	0.022716	0.001829	0.000453	0.247808	509.8853	7.0858	0.013897
[0.32, 0.49)	0.018594	0.000392	0.021087	0.002900	0.000720	0.248166	579.5645	7.1239	0.012292
[0.49, 0.63)	0.017444	0.000378	0.021680	0.003874	0.000957	0.247144	596.8228	6.8889	0.011543
[0.63, 0.77)	0.016789	0.000357	0.021237	0.003034	0.001294	0.426602	515.4993	6.2749	0.012173
[0.77, 0.95)	0.015105	0.000297	0.019683	0.003310	0.000768	0.232102	544.5999	7.2162	0.013250
[0.95, 1.18)	0.012398	0.000249	0.020057	0.004258	0.000933	0.219190	380.0781	6.1845	0.016272
[1.18, 1.80)	0.008227	0.000153	0.018561	0.003692	0.000794	0.215045	378.5408	5.2391	0.013840

LD_2 Target: Systematic Components (Table_Systematics_LD2.tex)

Latest Results

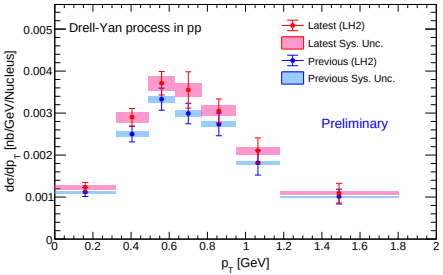
$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.018048	0.000557	0.030842	0.001829	0.000453	0.247808	614.7577	5.1141	0.008319
[0.32, 0.49)	0.017107	0.000492	0.028767	0.002900	0.000720	0.248166	727.0675	5.9278	0.008153
[0.49, 0.63)	0.016748	0.000486	0.029001	0.003874	0.000957	0.247144	688.8200	5.6655	0.008225
[0.63, 0.77)	0.015873	0.000454	0.028603	0.003034	0.001294	0.426602	641.4847	5.4729	0.008532
[0.77, 0.95)	0.014461	0.000381	0.026328	0.003310	0.000768	0.232102	673.2765	5.8236	0.008650
[0.95, 1.18)	0.012216	0.000323	0.026452	0.004258	0.000933	0.219190	547.6337	5.6667	0.010348
[1.18, 1.80)	0.008176	0.000199	0.024301	0.003692	0.000794	0.215045	510.6518	6.5158	0.012760

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.020298	0.000441	0.021727	0.001829	0.000453	0.247808	643.1022	7.4817	0.011634
[0.32, 0.49)	0.019066	0.000392	0.020565	0.002900	0.000720	0.248166	754.9459	8.3534	0.011065
[0.49, 0.63)	0.017807	0.000378	0.021237	0.003874	0.000957	0.247144	720.0996	8.1358	0.011298
[0.63, 0.77)	0.017477	0.000357	0.020401	0.003034	0.001294	0.426602	673.0926	7.1867	0.010677
[0.77, 0.95)	0.015424	0.000297	0.019276	0.003310	0.000768	0.232102	685.3519	8.1330	0.011867
[0.95, 1.18)	0.013143	0.000249	0.018919	0.004258	0.000933	0.219190	523.4376	6.5040	0.012426
[1.18, 1.80)	0.008446	0.000153	0.018081	0.003692	0.000794	0.215045	496.0850	5.4892	0.011065

Cross-Section Comparison vs. p_T: LH_2 and LD_2

LH_2 Plot



LD_2 Plot

